Studying temperature dependence of the Hall Coefficient in p-type semiconductors

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1 Abstract

The Hall effect results in a voltage difference across a conductor due to applying a magnetic field perpendicular to the current flow. This experiment involves instrument calibration and aims to measure the Hall coefficient. Additionally, we investigate the temperature-dependent behavior of the Hall coefficient to identify the inversion temperature indicating a transition in the dominant charge carrier type within the material.

2 Apparatus

- Hall probe (A single-crystal Ge probe with spring-type pressure contacts and four leads for current and Hall voltage measurements.)

- A compact, removable oven with a temperature range of ambient to 100°C, requiring 12W of power, facilitating precise temperature control.
- Chromel-Alumel thermocouple with its junction positioned 1 mm away from the crystal
- Hall effect set-up (DHE-22)
- An electromagnet (EMU-50V) for inducing the Hall effect which offers precise control over the magnetic field strength.
- Power Supply (DPS-50) to provide a stable and controllable current
- Digital Gaussmeter (DGM-102) to measure the strength of the magnetic field applied to the Hall Probe

3 Theory

Hall effect is based on Lorentz force experienced by charged particles moving in an electromagnetic field. The force acting on the charge carriers is represented as $\vec{F}_m = -e(\vec{v} \times \vec{B})$, and it is balanced by the force $\vec{F}_H$ due to the Hall field $\vec{E}_H$: $\vec{F}_H = -e\vec{E}_H$. Since the velocity $\vec{v}$ is aligned with the y-axis and the magnetic field $\vec{B}$ is along the z-axis, the electric field $\vec{E}$ emerges along the x-axis and can be expressed as:

$$-F_m/e = vB = \mu E_x B$$

where μ denotes carrier mobility defined by $v = \mu E_x$. Here, E_x represents the applied elec- tric field along the x-axis, and v signifies the drift velocity of the electrons. This electric field is related to the current density and conductivity by $\sigma E_x = J_x$. So, taking a metallic sample and passing current I through it with a magnetic field B per- pendicular to current flow, assuming thickness (the linear dimension parallel to the applied magnetic field) being t and conduction electron concentration as n, we have Hall voltage V_H across the direction orthogonal to both B and I as:

$$V_H = \frac{1}{t}\left(-\frac{1}{ne}\right) I \cdot B$$

Here for a metal, we define the Hall coefficient as:

$$R_H = -\frac{1}{ne}$$

and it is given in terms of V_H as:

$$R_H = \frac{V_H t}{IB}$$

For a semiconductor crystal, we have:

$$R_H = \frac{1}{e}\frac{\mu_h^2 p - \mu_e^2 n}{(\mu_h p + \mu_e n)^2}$$

where μ_h and μ_e are mobilities of holes and electrons, respectively.

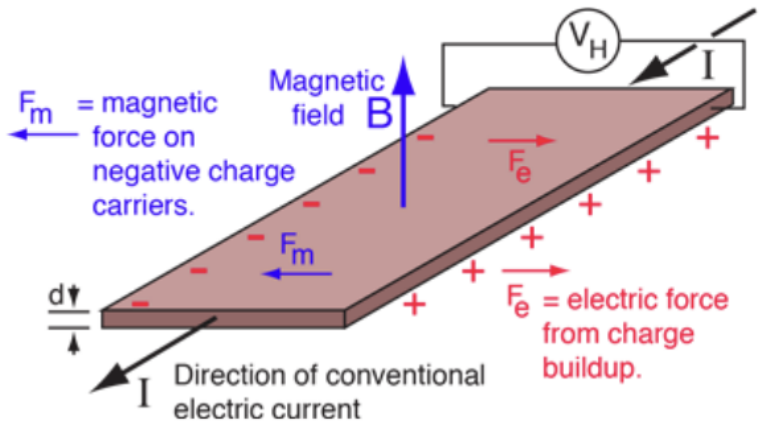


Figure 1: Hall effect

Notably, since the mobilities μ_h and μ_e are not constants but functions of temperature (T), the Hall coefficient is also temperature-dependent and may even change sign. Generally, $\mu_e > \mu_h$, which suggests at high temperatures the Hall coefficient may change its sign, re- ferred to as "Hall coefficient inversion," is a characteristic feature of "p-type" semiconductors exclusively.

4 Experimental set-up

The DHE-22 setup consists of two sub-units, each comprising two essential components. **(a) Measurement of Probe Current and Hall Voltage:** This unit includes a digital milli-voltmeter and a constant current power supply, facili- tating the simultaneous reading of the Hall voltage and probe current on the same digital panel meter using a selector switch.

- **Digital Milli-voltmeter:** This component uses an external reference to prevent per- formance degradation due to internal heating. It offers convenience in the Hall Exper- iment by accommodating input voltages of either polarity, with a range of 0-200 mV (100 μV minimum). This ensures the accuracy of our Hall voltage measurements.

- **Constant Current Supply:** Specifically designed for Hall Probe applications, this power supply ensures protection against crystal burn-out caused by excessive current. It provides highly regulated and nearly ripple-free DC output, guaranteeing the safety and reliability of our experimental setup.

(b) Measurement of Thermo EMF and Heater Current: This unit comprises a digital millivoltmeter and a constant current power supply, allowing measurement of the thermocouple's thermo EMF and heater current on the same Digital Panel Meter (DPM) using a selector switch.

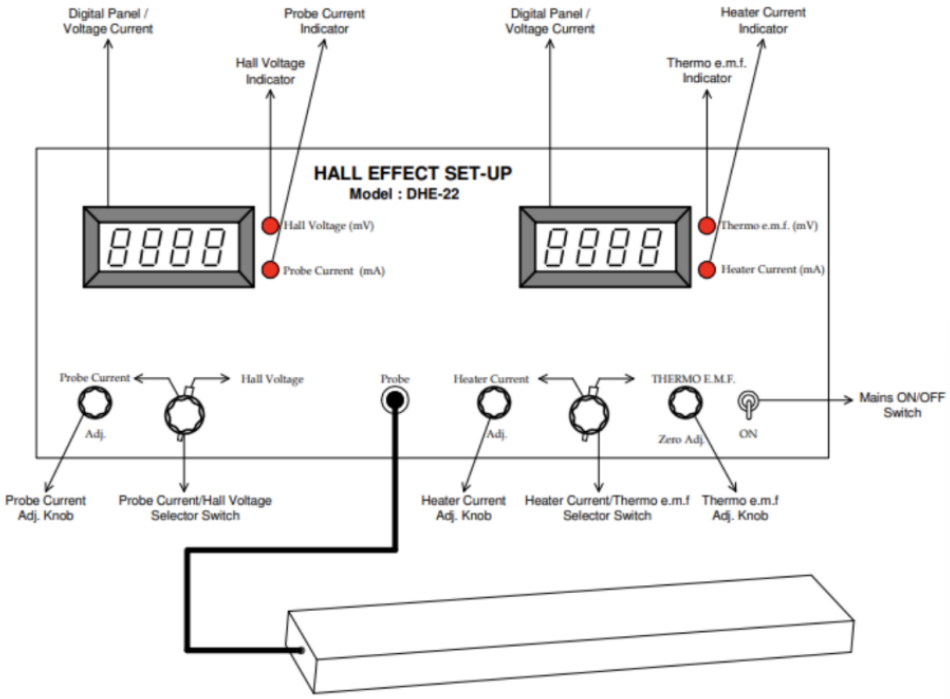


Figure 2: Hall effect setup

5 Observations & Data Analysis

5.1 Calibration factor determination

°C	0	1	2	3	4	5	6	7	8	9
0	0	0.04	0.08	0.12	0.16	0.2	0.24	0.28	0.32	0.36
10	0.4	0.44	0.48	0.52	0.56	0.6	0.64	0.68	0.72	0.76
20	0.8	0.84	0.88	0.92	0.96	1	1.04	1.08	1.12	1.16
30	1.2	1.24	1.28	1.32	1.36	1.4	1.44	1.49	1.53	1.57
40	1.61	1.65	1.69	1.73	1.77	1.81	1.85	1.9	1.94	1.98
50	2.02	2.06	2.1	2.14	2.18	2.23	2.27	2.31	2.35	2.39
60	2.43	2.47	2.51	2.56	2.6	2.64	2.68	2.72	2.76	2.8
70	2.85	2.89	2.93	2.97	3.01	3.05	3.1	3.14	3.18	3.22
80	3.26	3.3	3.35	3.39	3.43	3.47	3.51	3.56	3.6	3.64
90	3.68	3.72	3.76	3.81	3.85	3.89	3.93	3.97	4.01	4.06
100	4.1	4.14	4.18	4.22	4.26	4.31	4.35	4.39	4.43	4.47
110	4.51	4.55	4.6	4.64	4.68	4.72	4.76	4.8	4.84	4.88

Figure 3: Calibration table for Chromel-Alumel

Current in A	Magnetic field in Gauss	Current in A	Magnetic field in Gauss
0.1	80	2.1	1800
0.2	150	2.2	1880
0.3	230	2.3	1960
0.4	320	2.4	2040
0.5	400	2.5	2130
0.6	490	2.6	2220
0.7	580	2.7	2300
0.8	660	2.8	2380
0.9	750	2.9	2480
1	830	3	2560
1.1	920	3.1	2640
1.2	1010	3.2	2720
1.3	1090	3.3	2810
1.4	1180	3.4	2900
1.5	1270	3.5	2970
1.6	1360	3.6	3050
1.7	1450	3.7	3130
1.8	1530	3.8	3220
1.9	1620	3.9	3290
2	1700	4	3370

Table 1: Data for calibration

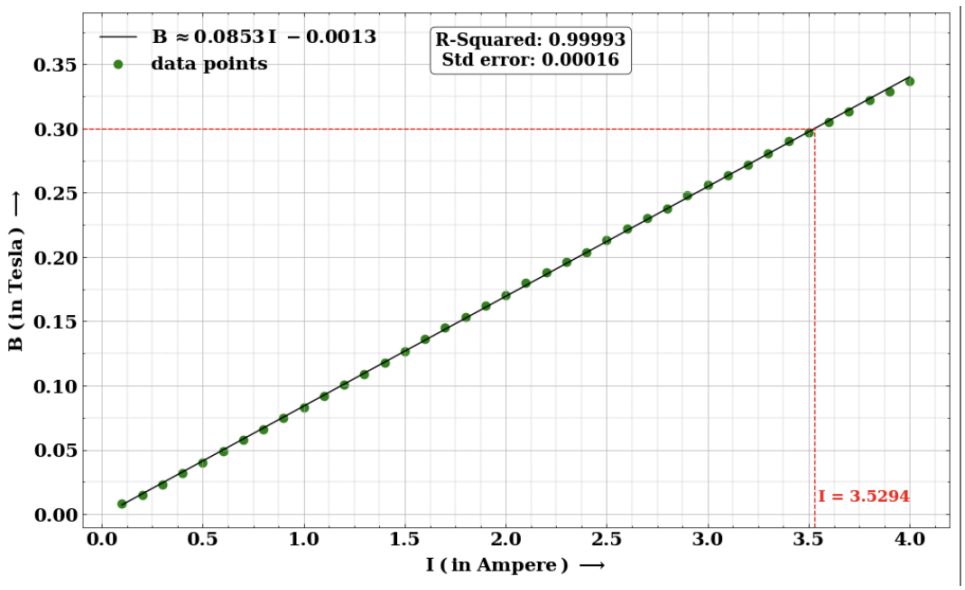


Figure 4: Calibration data with best-fit line

5.2 Inversion temperature determination

Given sample thickness $t = 0.5$ mm. The following readings are noted keeping probe current, $I = 4.01$ mA, $B = 0.3$ T with current in electromagnet = 3.5294 A (determined from the calibration curve).

Heater Current (mA)	Thermo EMF (mV)	Temperature (°C)	Hall Voltage (mV)	Offset Voltage (mV)	Corrected Hall Voltage (mV)
0	0	30.5	46.2	2.6	43.6
200	0.12	33.5	45.1	2.7	42.4
300	0.29	37.25	43.4	2.1	41.3
400	0.62	46	39.4	1.9	37.5
500	0.9	53	33.5	2.5	31
600	1.47	67.1	19.2	2.1	17.1
650	1.83	76	7.6	1.1	6.5
700	2.02	80.5	5.2	1.4	3.8
750	2.2	85.1	3.5	1.6	1.9
800	2.45	91	0.6	1.1	-0.5
850	2.7	97	-0.8	1	-1.8
900	3.09	106.3	-2.2	0.5	-2.7
950	3.37	113	-2.5	0.2	-2.7

Table 2: Data taken to plot Hall coefficient vs. Temperature

Temperature (°C)	Corrected Hall Voltage (mV)	Hall Coefficient (m^3C^{-1})	Carrier Concentration (in $10^{20}/m^3$)
30.5	43.6	0.0181	3.449
33.5	42.4	0.0176	3.5466
37.25	41.3	0.0172	3.6410
46	37.5	0.0156	4.01
53	31	0.0129	4.8508
67.1	17.1	0.0071	8.7938
76	6.5	0.0027	23.1346
80.5	3.8	0.0016	39.5724
85.1	1.9	0.0008	79.1447
91	-0.5	-0.0002	-300.75
97	-1.8	-0.0007	-83.5417
106.3	-2.7	-0.0011	-55.6944
113	-2.7	-0.0011	-55.6944

Table 3: Variation of Hall coefficient and carrier concentration with temperature

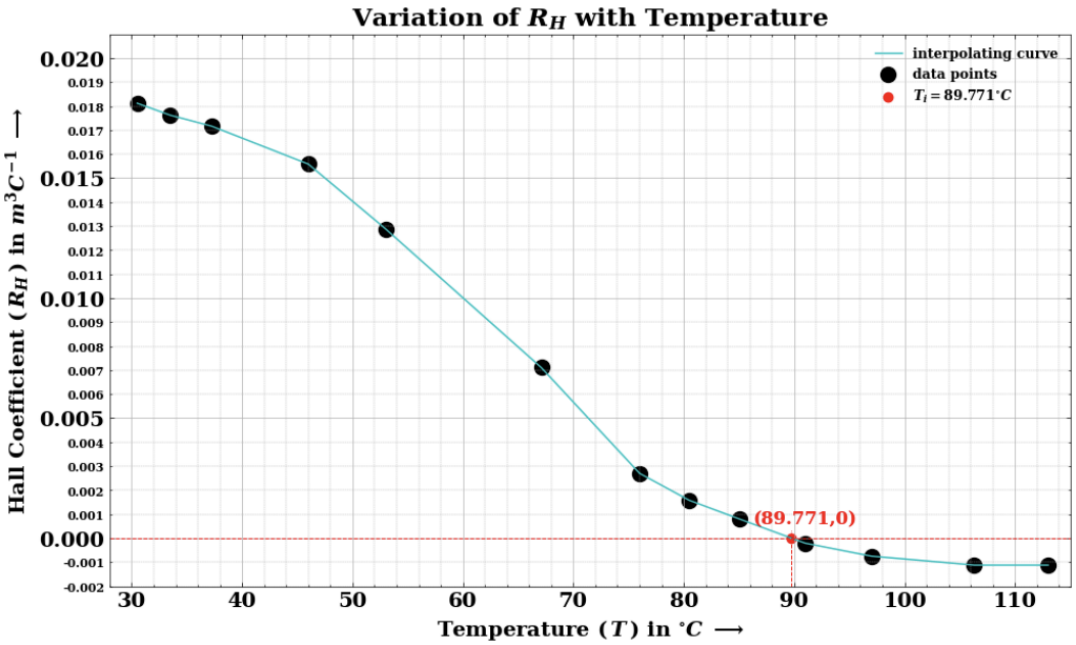


Figure 5: Hall Coefficient vs. Temperature of given Ge p-type sample

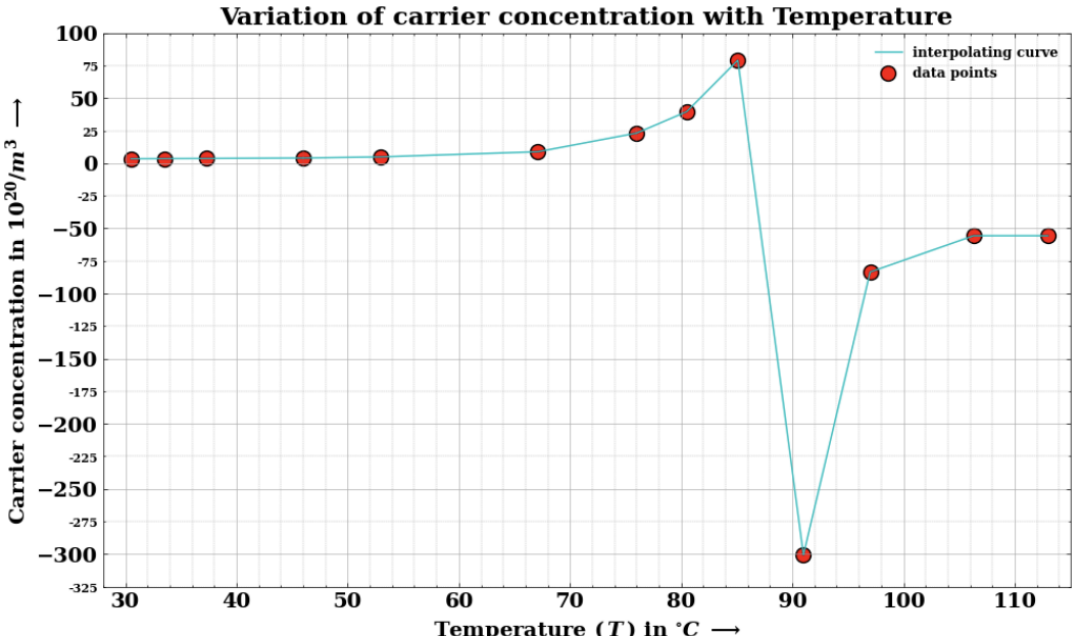


Figure 6: Carrier concentration vs. Temperature of given Ge p-type sample

6 Error Analysis

6.1 Possible sources of Error

The possible sources of errors that can occur during the experiment:

- Impurities and compositional variations in semiconductor materials can introduce er-

- The semiconductor material, as well as the Hall probe, can be affected by fluctuations in temperature.
- The precise positioning of the Hall probe within the magnetic field is critical.
- Non-uniformity in the applied magnetic field can lead to inaccuracies in Hall voltage measurements.
- Electrical noise in the measurement equipment can introduce errors in voltage and current readings.
- The quality of the electrical contacts with the semiconductor material can impact the accuracy of current measurements.
- Incorrect calibration of instruments used for measuring voltage, current, or magnetic field strength can introduce systematic errors in the data.

6.2 Error in calibration factor

For the calibration factor, we have:

$$K_{calib} = \frac{G}{I}$$

where G is the field and I is the current. So, we have (excluding numerical errors in fitting the best-fit line):

$$\frac{\delta K_{calib}}{K_{calib}} = \frac{\delta G}{G} + \frac{\delta I}{I} \approx \frac{10}{3000} + \frac{0.01}{3.5294} \approx 0.0062 = 0.62\%$$

So, we have an approximate error of 0.62 % for the calibration coefficient determination.

6.3 Error in Hall coefficient and inversion temperature

From (3), we have:

$$\frac{\delta R_H}{R_H} = \frac{\delta V_H}{V_H} + \frac{\delta t}{t} + \frac{\delta I}{I} + \frac{\delta B}{B}$$

Since the thickness value was given, we assume $\delta t = 0$. Now, $\delta V_H = 0.1$ mV and taking the middle reading of $V_H = 17.1$ mV in the error calcu- lation. δI for the probe current is 0.01 mA, and $\delta B = 10$ G as $\times 10$ Gauss probe was used for a field reading of 3000 G.

7 Results and Inference

Hence, on average, we can say that the relative error in Hall coefficient determination for this experiment is:

$$\left(\frac{\delta R_H}{R_H}\right)_{avg} = \frac{0.1}{17.1} + \frac{0.01}{4.01} + \frac{10}{3000} \approx 0.0117 = 1.17\%$$

Now, to find the error in the determination of the inversion temperature, we have to also take into account the numerical error associated with the interpolating function.

However, the principal error contribution will be from the relative error in Hall coefficient determina- tion along with that of temperature determination. Hence, very roughly we can estimate the relative error for the inversion temperature as:

$$\frac{\delta T_{inv}}{T_{inv}} \sim \frac{\delta R_H}{R_H} + \frac{\delta T}{T}$$

Now, $\delta T = 0.25°\text{C}$ and $T_{inv} \approx 89.771°\text{C}$. So, we have roughly:

$$\frac{\delta T_{inv}}{T_{inv}} \sim 0.0117 + \frac{0.25}{89.771} \approx 0.0145 = 1.45\%$$

Thus, approximately we have an error of 1.45 % for the inversion temperature determination.

Results and Inference

From the data analysis, we can conclude that:

- The calibration coefficient for the given electromagnet is determined to be $K_{calib} \approx 853 \pm 5.289$ G/A
- The Hall coefficient of the given p-type semiconductor showed a decrease with an increase in temperature and eventually underwent an inversion.
- The phenomenon of Hall coefficient inversion was observed. We report the inversion temperature as $T_{inv} \approx 89.771 \pm 1.302°\text{C}$.

From the results, we can draw the following inference:

- The observation of a linear relationship between the magnetic field and current, along with low error, confirms the proper calibration of the instrument and validates the experimental results.
- The sign change in the Hall coefficient at the inversion temperature indicates the transition from dominant electron carriers to dominant hole carriers in p-type semiconductors. This is also confirmed by the carrier concentration plot.